Resilience and Cost in Syntactic Prediction: Evidence from Either–Or Constructions

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Prediction is a fundamental top-down process during language processing, allowing readers to anticipate not only upcoming words but also syntactic structures to enable efficient processing (Ferreira & Qiu, 2021; Pickering & Gambi, 2018). Either-or constructions provide a valuable tool for investigating syntactic prediction during reading (Staub & Clifton, 2006). When readers encounter the word *either* in a sentence, they might anticipate a following *or*-construction, and a facilitation effect will be observed when the *or*-construction eventually appears.

The present eye-tracking study investigated whether syntactic prediction can be maintained across intervening text, and whether a processing cost is obtained when the intervening text conflicts with syntactic prediction. We manipulated the presence of the word *either* and the location of the intervening *and*-construction, which appeared either *before* or *after* the anticipated *or*-construction (see example stimulus in *Figure* 1). 80 experimental sentences and 60 filler sentences were adopted in this study.

36 native English speakers took part in this eye-tracking study. The analysis was undertaken using general linear mixed-effects models in R on the measurements of *gaze duration* (GD) and *regression path duration* (RPD) at *and*- and *or*-constructions. We observed an interaction between the variables of *either-presence* and *and*-construction location at the region of the *and*-construction on *gaze duration* (*Est* = 72.61, *SE* = 17.70, $t = 4.10$, $p < .001$) and *regression path duration* (*Est* = 73.61, *SE* = 28.81, $t = 2.56$, $p = .01$) (see *Figure* 2 **Panel A**). There were longer fixation durations when *either* is present and the *and*-construction takes place before the *or*-construction. A resilient facilitation on the *or*-construction was also found on *gaze duration* (*Est* = 38.81, *SE* = 10.75, $t = 3.61$, $p < .001$) and *regression path duration* (*Est* = 30.71, *SE* = 12.29, $t = 2.50$, $p = .01$) when *either* is present compared to absent regardless of the location of the intervening *and*-construction (see *Figure* 2 **Panel B**). While no significant interaction between the two variables is observed (GD: *Est* = -24.18, *SE* = 19.13, $t = -1.26$, $p = .21$; RPD: *Est* = -39.42, *SE* = 33.22, $t = -1.19$, $p = .24$).

In sum, the results indicate that syntactic expectations can be maintained across intervening material, though local processing difficulty arises when unexpected structures disrupt anticipation.

Key Words: Syntactic Prediction; Syntactic Processing; Reading; Eye-tracking

References:

Ferreira, F., & Qiu, Z. (2021). Predicting syntactic structure. *Brain Research*.

Pickering, M. & Gambi, C. (2018). Predicting while comprehending language: A theory and review. *Psychological Bulletin*.

Staub, A., & Clifton, C. (2006). Syntactic prediction in language comprehension: Evidence from either...or. *Journal of Experimental Psychology: Learning, Memory, and Cognition*.

Figure 1. The example stimulus illustrates the manipulation of *either-presence* (present / absent) and the location of the intervening *and*-construction (before / after *or*-construction).

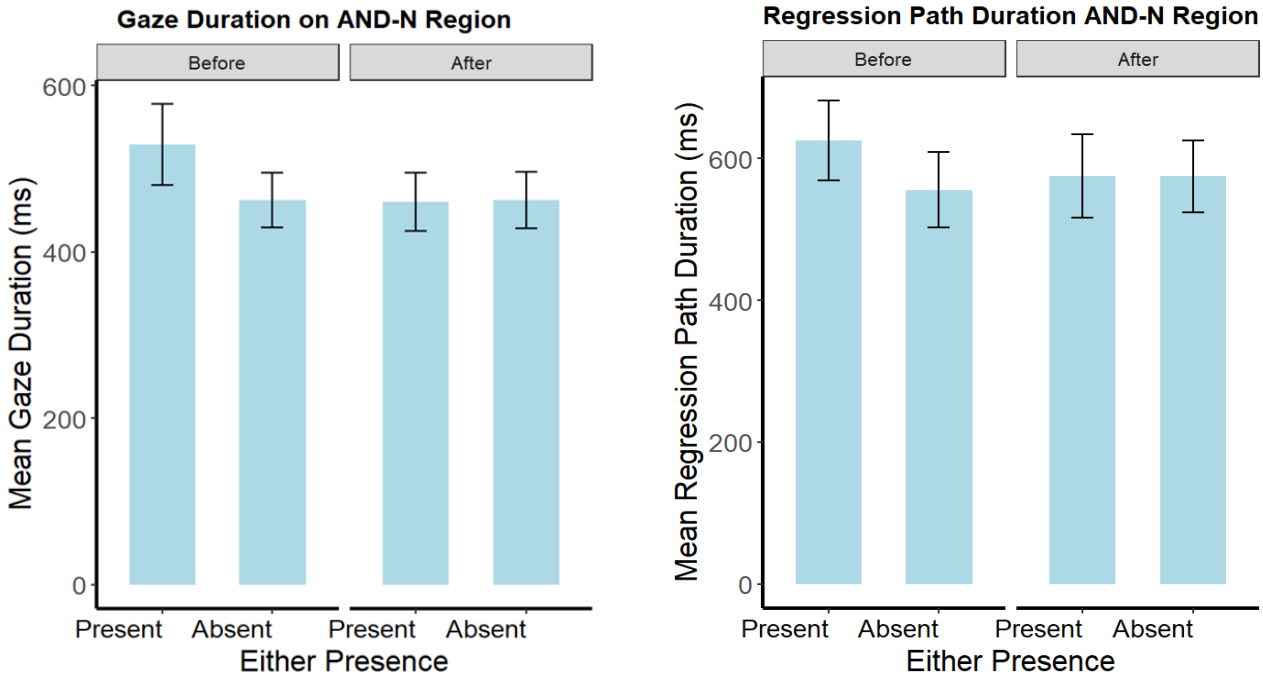
She likes to have **either** muesli and fruit or yoghurt for her breakfast.
either-N and-N or-N (present / before)

She likes to have muesli and fruit or yoghurt for her breakfast.
either-N and-N or-N (absent / before)

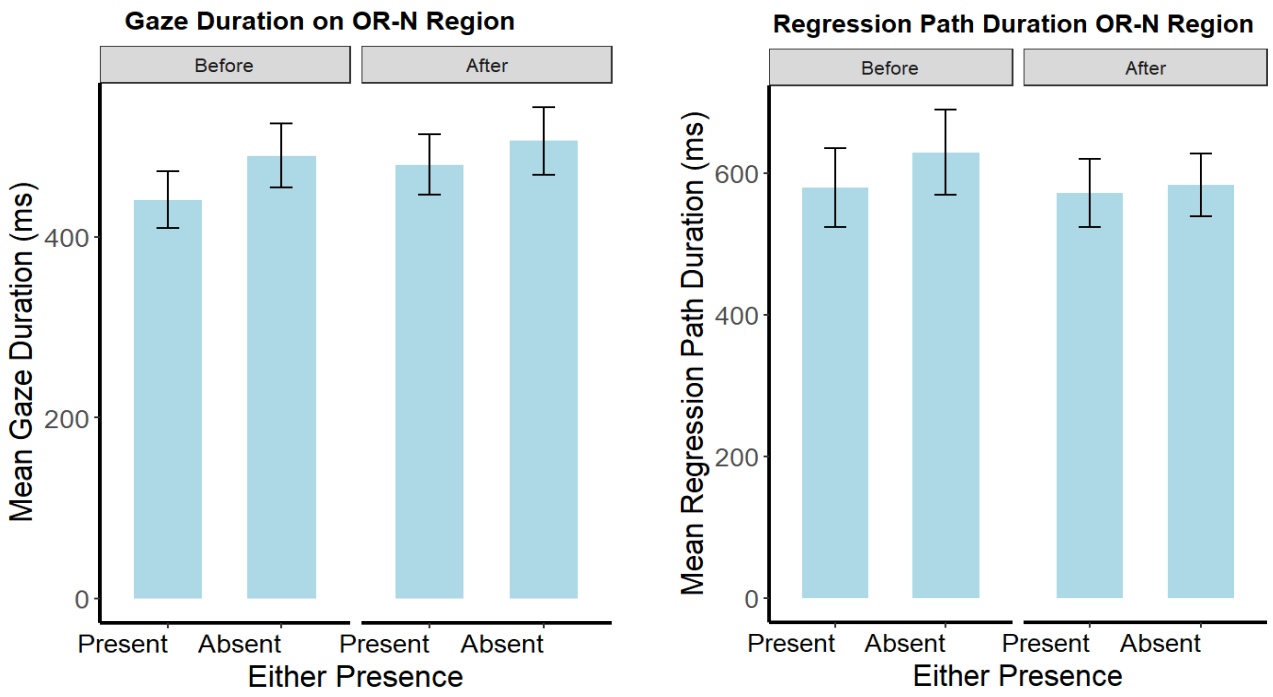
She likes to have **either** muesli or yoghurt and fruit for her breakfast.
either-N or-N and-N (present / after)

She likes to have muesli or yoghurt and fruit for her breakfast.
either-N or-N and-N (absent / after)

Figure 2. Means ± 95% confidence intervals for Gaze Duration and Regression Path Duration on ***and***-construction Region and ***or***-construction Region.



Panel A


Gaze Duration on OR-N Region
Before
After
Mean Gaze Duration (ms)
0
200
400
Present
Absent
Present
Absent
Either Presence
Regression Path Duration OR-N Region
Before
After
Mean Regression Path Duration (ms)
0
200
400
600
Present
Absent
Present
Absent
Either Presence

Panel B